Process and Plant Design for Fish Skin Collagen

Material grade feedstock for films and coatings

Design Revision 021
28 October 2026
Issued for project review

Design basis: 200 tonnes per year of contained collagen. Native Washing and Dialysis Aspen designs remain fixed. Physical equipment, operating schedules and mechanical services are coordinated externally.

Contents

1 Executive summary **3**

2 Document control and design authority **3**

3 Introduction and project objectives **3**

4 Common design basis **4**

4.1 Material representation and assumptions 4

5 Product requirements and route comparison **5**

6 Operating campaign and capacity **5**

6.1 Discrete freeze-dryer loading 6

7 Equipment selection and thermal design **6**

7.1 Vessel capacity and conditioning 6

7.2 Individual cooler design 7

8 Piping and pump selection **8**

9 Utilities cleaning and collection **9**

9.1 Selected CIP programme and balances 9

9.2 Collection receiver capacities 10

10 Operation control and physical arrangement **10**

11 Capital reconciliation and economic scenarios **11**

12 Verification and remaining closeout **12**

13 References and traceability **13**

A Master equipment and service register **14**

B Individual equipment data sheets **16**

B.1 T-101A/B Alkaline pretreatment and skin rinsing 16

B.2 T-102 Acetic acid preparation 17

B.3 T-201A-F Collagen extraction 17

B.4 T-301A-E Salting and cooling 18

B.5 S-101 Raw skin mechanical preparation 18

B.6 S-201 Extraction clarification 19

B.7 S-301 Initial salt separation . . . 19
B.8 F-301A01-A14 and B01-B14 Eight-cycle cake reslurry washing . . . 19
B.9 T-501A/B Metered cold wash-water storage . . . 20
B.10 E-501 Wash-water cooler . . . 20
B.11 E-401 Redissolved collagen cooler . . . 21
B.12 E-402 Dialysis bath make-up cooler . . . 21
B.13 P-701 Alkaline waste . . . 22
B.14 P-101 Alkali feed . . . 22
B.15 P-501A/B Cold wash water . . . 23
B.16 P-103 Extraction acid . . . 23
B.17 P-102 Fresh rinse water . . . 24
B.18 P-201A-C Product slurry transfer . . . 24
B.19 P-500 Wash water cooling preparation . . . 25
B.20 PG-501 E501 glycol branch circulation . . . 25
B.21 PG-301 Salting coil glycol circulation . . . 25
B.22 PH-201 Extraction hot-water circulation . . . 26
B.23 CH-601 Process-only glycol cooling . . . 26
B.24 HW-602 Extraction hot-water service . . . 26
B.25 CIP-801 Cleaning service availability . . . 27
B.26 TK-601 Process glycol hydraulic buffer . . . 27
B.27 EV-601 Glycol expansion vessel . . . 28
B.28 TK-602 Extraction hot-water buffer . . . 28
B.29 TK-801 Active CIP alkaline stock . . . 28
B.30 T-701 Alkaline and selected CIP collection . . . 29
B.31 T-703 Combined acid salt and wash-filtrate collection . . . 29
B.32 T-705 FD condensate collection . . . 29
B.33 HT-801 CIP initial stock and makeup heating . . . 30
B.34 Freeze dryer individual cabinet loading . . . 30

1 Executive summary

This revision coordinates the frozen Washing process at a200 t/y contained-collagen design case. The physical lot is 18 times the 270.27027 kg native wet-skin batch. The declared schedule delivers 1,145 whole lots, about 200.10t contained collagen and 214.20t powder, during a7920 h process campaign. The first product is released about 133.8 h after the first fish-skin charge.

The principal equipment correction is discrete dryer loading. Each physical lot requires 288 m2 active shelves at the accepted loading and cycle, supplied by two 136 m2 cabinets and one 16 m2 cabinet. Seven independent groups require 21 cabinets in total. The salting service requires five vessels after the selected cooling time is included. Two 14-module wash banks preserve 216 m2 active filter area per lot.

The process PG loop peaks at 488.183 kW thermal; the selected process-only allowance is 600 kW. Extraction hot water peaks at 108.737 kW, with a140 kW heater allowance. These are separate from complete freeze-dryer packages and the selected CIP utility. Twenty-three line cases and actual published pump curves support the mechanical service review.

The capital reconciliation retains the inherited unquoted EUR method and changes equipment counts and package assumptions. It gives a purchase-budget scenario of EUR 28.960 million and fixed capital of EUR 105.125 million. It is an equipment-count reconciliation, with utility/package, site and staff scope still requiring the closing actions in this report. EUR 250/kg powder remains a sales-price scenario.

2 Document control and design authority

Record	Revision or purpose
COLL-PDB-021	Common process and plant design basis; this report
COLL-MEB-021	V009 native stream readback and V021 physical stream inventory
COLL-SCH-021	V021 resource-constrained operating campaign and independent checks
COLL-MEC-021	V021 hydraulic, thermal, pump and utility calculations
COLL-DS-021	Current equipment register and service data sheets
COLL-DRG-021	Drawing revision pending Spyder execution; BFD/PFD assigned separately
Issue purpose	Project review; original native Aspen and historical calculations preserved

The report uses one equipment and service register, a common pressure and time convention, and explicit interfaces between systems. NREL/Harris provides the process-area and calculation-appendix model; the Shell Peterhead design package provides the common-basis, identification, utility and interface model. These are organizational references rather than sources of fish-collagen operating numbers.

3 Introduction and project objectives

Fish skin is a potential collagen feedstock generated alongside fish processing. The intended product in this study is a raw material for films and coatings. Nilsuwan et al. investigated films made from salmon-skin acid-soluble collagen solutions and showed that processing conditions affected material properties. Their work supports the application choice while also showing why composition alone cannot establish film performance [1].

Published lumpfish work provides preparation and extraction context. Scheja et al. investigated alkaline pretreatment and acetic-acid extraction and reported lower recovery in their scale-up ex-

periment than under optimized laboratory conditions [2]. Yield denominators and scale assumptions must therefore remain visible throughout a plant design. The present design uses a declared literature-based recipe and material representation.

The project objective is to compare a reference purification process with a simplified material-grade route, quantify resource and equipment consequences, and identify the conditions supporting an engineering recommendation. Dialysis followed by freeze drying is the reference. Cake washing after precipitation, followed by freeze drying, is the detailed engineering case. They are alternative designs; the Washing plant does not install the reference route as a second product line.

The scope includes mass and energy accounting, batch capacity, main equipment, piping and pumping, heat transfer, utilities, cleaning and waste interfaces, control, site/layout premises and economic scenarios. Detailed calculations are supplied in the new Excel calculation book and reproducible scripts. Formal text is maintained in LaTeX. BFD and PFD drawings will be incorporated from the separately assigned drawing team.

4 Common design basis

Quantity	Value	Meaning
Wet skin per native batch	270.270 kg	Normalized Aspen inventory
Purification collagen inlet	10.000 kg/native batch	Recovery screen denominator
Physical multiple	18	Same recipe, T, P and component endpoints
Wet skin per physical lot	4,864.865 kg	External equipment basis
Whole lots delivered	1,145/y	No product opening or closing WIP
Contained collagen output	200.097588 t/y	Main production basis
Powder output	214.200684 t/y	Sales mass differs from contained collagen
Process campaign	7920 h	330 days including process startup and rundown
Cold-year utility preparation	2 h before first feed	CIPstock heating is not a free initial condition
Cold-year final cleanup	Calendar 7922 h	838 h remain in the 8760 h calendar
Source pressure	100000 Pa absolute	1.0 bar absolute, not 1 barg or 101325 Pa

One numerical Aspen hour carries one native batch inventory. Physical hourly flows are derived from transfer windows, while energy per batch is divided by a declared operation duration to obtain duty. The saved Washing_0.4 and Dialysis_0.4 files are not rerun or altered in this revision. V017/V018 extraction, rinse, hold-up and dryer-performance improvement targets are excluded from the adopted process.

4.1 Material representation and assumptions

Water/acetic-acid properties use the saved molecular NRTL-HOC setup; collagen and OTHERDRY are nonconventional bookkeeping materials, and acid/base/salt equivalents are decoded for accounting. The representation supports approximate component and enthalpy accounts. It does not predict electrolyte pH, real salt-out equilibrium, molecular integrity or film performance. Low-temperature parameter extrapolation and the accepted Newtonian property proxies remain declared course assumptions.

The washing calculation retains eight 15 min contacts and the original pressure/resistance/release scenario. Real wash charging adds 0.212842 h per physical lot to equipment handling time. The component endpoint remains fixed during this added handling allowance; no new fish-collagen kinetic parameter has been inferred.

5 Product requirements and route comparison

Route	Requirement	Project limit	Result	Basis
Washing low loss	Collagen fraction of nonvolatile material	>=0.98	0.983326	kg/kg nonvolatile
Washing low loss	Powder water fraction	<=0.05	0.049991	kg/kg powder
Washing low loss	NaCl fraction of nonvolatile material	<=0.001	0.000435	kg/kg nonvolatile
Washing low loss	Residual acetic acid per contained collagen	<=0.001	0.000009	kg/kg collagen
Washing low loss	Purification collagen recovery	>=0.95	0.970876	kg/kg purification inlet collagen
Dialysis	Collagen fraction of nonvolatile material	>=0.98	0.986302	kg/kg nonvolatile
Dialysis	Powder water fraction	<=0.05	0.049825	kg/kg powder
Dialysis	NaCl fraction of nonvolatile material	<=0.001	0.000570	kg/kg nonvolatile
Dialysis	Residual acetic acid per contained collagen	<=0.001	0.000187	kg/kg collagen
Dialysis	Purification collagen recovery	>=0.95	0.980537	kg/kg purification inlet collagen

These inherited project screens evaluate the saved component endpoints. Film strength, elongation, barrier performance, molecular integrity and storage stability remain application qualification requirements; numerical customer limits have not been assigned.

Route endpoints and normalized resource intensities	Washing	Dialysis
Collagen kg/native batch	9.708762	9.805371
Process fresh water m3/kg collagen	1.160829	3.239254
Dryer feed water kg/kg collagen	5.739	102.808
Native-equivalent active shelves m2	16	320
Accepted total FD cycle h	43.917873	47.462881

Reference E401 and E402 thermal cases are calculated at 18 times the native reference batch for service comparison. They are not installed in the Washing plant. This revision does not claim a completed lot-dispatch design or final 200 t/y investment package for the Dialysis reference. The final same-scale full-capital comparison remains a separate closeout item.

6 Operating campaign and capacity

For physical-lot collagen output m_C, target M_C and whole-lot count N,

$$N = \lceil M_C/m_C \rceil , \tag{1}$$

$$\Delta t = \frac{H_{\text{campaign}} - t_{\text{lot,clear}}}{N-1}. \tag{2}$$

Here $t_{\text{lot,clear}}$ is the first lot's complete equipment clearance time relative to its own start, including cleaning and turnaround.

Resource	Parallel positions	Minimum occupancy h	Actual max h	Reuse idle h
PREP	1	2.500	2.500	4.305
ALKW	2	13.508	13.508	0.102
EXTR	6	39.940	39.940	0.892
CLAR	1	2.964	2.988	3.842
SALT	5	28.453	28.971	5.573
CENT	1	3.011	3.106	3.794
WASH	2	12.956	12.956	0.655
FD	7	43.918	43.918	3.719
PACK	1	1.750	1.750	5.055
ACID_PREP	1	3.159	3.686	3.646
COLD_BUFFER	2	13.137	13.137	0.474
WATER_COOLER	1	2.000	2.000	4.805

The launch spacing is 6.805266635 h. The explicit event dispatch checks equipment, compatible transfer channels, cleaning deadlines and cabinet reservations. An initial 21.990665 m3 alkaline drain at 90 m3/h was added after the 6 h alkali contact; this outlet is separate from the six skin-rinse drains. Four cleaning circuits are feasible under the declared dispatch: one FD group circuit and three general circuits. A three-circuit trial failed a cleaning deadline.

The deterministic schedule reserves 13740 equipment intervals,29770 pump jobs,14885 CIP services and 3435 cabinet jobs. Independent conflict/deadline checks pass for the declared resources. The alkaline positions are tight, with about 0.102 h nominal reuse clearance. Unplanned outages and an operator shift roster are not included in this deterministic capacity claim.

6.1 Discrete freeze-dryer loading

The required area per physical lot is $18 \times 16 = 288$ m^2. Each group contains $136 + 136 + 16 = 288$ m^2, and seven groups provide 2016 m^2 across21cabinets. Large cabinets carry8.5native batches each, with25.5kg/h primary vapour capacity; the small cabinet carries one native batch with3kg/h capacity. The unchanged43.917873 h cycle already includes3hturnaround and1hpumpdown; these allowances are not added again.

CIP stock is heated during calendar 0–2 h before first feed. Archived process times use the first raw feed as time zero. Shifting them by 2 h gives cold-year final cleanup at 7922 h. In repeating years the utility preparation occupies the final 2 h of the nonproduction window. No product-lot pooling buffer is needed for the dedicated dryer groups; two cold-water buffers remain installed.

7 Equipment selection and thermal design

7.1 Vessel capacity and conditioning

The nominal-volume check includes coil displacement,

$$V_{\mathrm{nominal}} \geq \frac{m/\rho + V_{\mathrm{coil}}}{f_{\mathrm{working}}}, \qquad V_{\mathrm{coil}} = \frac{\pi D_o^2 L}{4}. \tag{3}$$

The working fraction is0.80 and the trial bulk density is1000kg/m3.

Vessel group	Quantity	Liquid m3/lot	Coil displacement m3	Nominal m3
T-101A/B	2	29.189	0.000	45
T-102	1	14.308	0.000	18
T-201A-F	6	21.463	0.908	36
T-301A-E	5	23.486	0.993	36

S101 is the mechanical raw-skin preparation service. Native DRAIN and SKWASH separation functions are represented by retaining and draining a skin basket in T101; the register does not require a second ordinary liquid pump for whole skins. Initial salt separation remains a separate service from the repeated wash-filter operations.

For finite utility circulation, with product thermal capacity C, utility capacity rate C_u and UA,

$$K_{\text{eff}} = C_u \left[1 - \exp(-UA/C_u)\right], \tag{4}$$

$$t = \frac{C}{K_{\text{eff}}} \ln \left| \frac{T_0 - T_u}{T_1 - T_u} \right|. \tag{5}$$

Service	Total area m2	Additional coil m2	Utility flow m3/h	Time h	Initial kW
J-201	66.039	40.840	36.000	1.792	108.737
J-301	72.252	44.683	144.000	2.217	317.553

A geometrically enlarged jacket alone does not retain the native conditioning time. Additional coil area supplies the required effective area. The displacement is included in vessel volume. OverallU=200W/m2/K remains an inherited engineering assumption rather than measured slurry heat-transfer performance.

7.2 Individual cooler design

The thermal method uses $Q = UA\Delta T_{\text{lm}}$ for true countercurrent paths and

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}. \tag{6}$$

Cooler	Route	Duty kW	Paths	Area m2	Process DP kPa	PG DP kPa
E-501	Washing	170.630	18	24.429	23.042	210.914
E-401	Dialysis	156.223	18	27.483	27.982	201.766
E-402	Dialysis	1,102.258	108	146.574	26.365	239.840

Each entry is one assembled parallel annular bank, not one purchased exchanger per flow path. Per-path geometry and flow preserve the recorded Reynolds numbers, resistance-basedU and pressure loss. A plate exchanger is credible for clean wash water, but this revision adopts the fully calculated annular-bank basis until an equivalent-duty plate rating supplies both-side pressure losses. E401 remains a protein-solution reference service.

8 Piping and pump selection

The hydraulic calculation uses Darcy friction and separately counted fittings,

$$\Delta P = \left(f_D L/D + \sum K\right) \rho v^2/2 + \Delta P_{\text{equipment}}, \tag{7}$$

$$H = \Delta P/(\rho g) + \Delta z, \tag{8}$$

$$\text{NPSH}_a = (P_{\text{source}} - P_v)/(\rho g) + z_{\text{surface}} - z_{\text{pump}} - h_{\text{suction}}. \tag{9}$$

The selected pressure boundary is 100000 Pa absolute. Minimum source surface, pump centre and maximum receiving surface are 1.2m,0.6m and 4.2m. The minimum source surface denotes an elevated draining low point, not an unreported retained heel. All pipe, pump and NPSH calculations use these same datums. Main transfer capacity is 90 m3/h; a line capacity does not by itself prescribe the selected brine-discharge timing.

Line	Flow m3/h	DN	Length m	Selected H m	NPSHa m
L101	90.000	150	20.200	4.500	10.495
L102	90.000	150	27.200	5.000	10.495
L103	90.000	150	39.200	5.000	10.070
L104	90.000	150	20.200	4.500	10.067
L106	90.000	150	38.200	5.000	10.067
L107	90.000	150	20.200	4.500	10.065
L108	90.000	150	36.800	2.000	9.028
L109	90.000	150	12.200	7.000	10.495
L110	90.000	150	77.800	2.500	10.495
L112	90.000	150	39.800	2.000	10.495
L113	90.000	150	39.800	2.000	10.495
L201	34.585	100	25.200	4.500	10.379
L202	9.519	50	23.200	8.500	10.027
L203	61.974	125	20.200	8.000	10.335
L204	74.511	150	27.800	1.500	10.136
L205	9.164	50	14.200	5.000	10.046
L500	9.578	50	14.200	7.500	10.305
G501	78.087	125	189.400	36.000	10.192
G401	71.493	125	151.400	33.000	10.219
G402	504.435	300	153.400	36.000	10.207
G301	144.000	200	113.400	9.500	10.179
H201	36.000	100	125.400	9.500	10.310
C701	110.000	150	169.400	33.000	6.743

L109 includes a conservative 40m DN150 full-flow header and a12m DN40 worst branch in the 14-module filter bank. Its selected head is 7.0m. Header and branch priming inventory is about 1.194 m3 per bank. Campaign priming and terminal drainage are separate inventory entries. Filtration gas differential 100kPa is excluded from the fill-pump head.

After every bank cleaning, warm/caustic header liquid is drained and the final cold rinse/refill uses the existing 2.26923 m3 wash-group rinse allowance. About 1.19393 m3 remains in the header and about 1.07530 m3 is available for flushing. The native wash-dose totalizer starts after the 4 C inlet temperature permissive. Repeated opening/closing header inventory cancels per lot. Campaign startup/final water adds 2.38786 m3 to the separate water ledger. The ability to achieve the cleaning/cooling requirement within 75 min remains part of CIP qualification.

Service	Published candidate	Duty flow m3/h	Required H m	Required drive kW
L109	LKH60/200 mm/1425 rpm	90.000	7.000	4
L500	LKH5/110 mm/2870 rpm	9.578	7.500	1.5
G501	LKH40/220 mm/2940 rpm	78.087	36.000	30
G401	LKH40/210 mm/2940 rpm	71.493	33.000	22
G301	LKH60/160 mm/2955 rpm	144.000	9.500	15
H201	LKH40/200 mm/1425 rpm	36.000	9.500	4
C701	LKH60/200 mm/2955 rpm	110.000	33.000	30

Ordinary 90 m3/h transfer services use the published LKH60/190 mm/1425 rpm curve, with a4 kW drive requirement. L109 requires the 200 mm low-speed curve after the manifold is included. Catalogue H, shaft power and NPSHr were extracted from the original Alfa Laval PDF vector paths for water 20 C. The stated viscosity screening factors and drive margins are project assumptions. G501 requires 30 kW by the adopted screen although the water catalogue curve lists 22 kW; the larger drive remains a procurement requirement.

Manufacturer minimum continuous flow, seals/materials and final viscous-service performance remain procurement qualifications. Reference-route G402 remains a high-flow hydraulic duty. The 23 line cases map to nine compatible main transfer channels, separate utility circulation pumps and the four CIP circuits.

9 Utilities cleaning and collection

Utility boundary	Calculated peak	Selected provision	Annual thermal quantity
Process PG 30 at-2 C	488.183 kW	600 kW process chiller	834,163.573 kWh
Extraction hot water 25 C	108.737 kW	140 kW heater	148,627.037 kWh
Primary FD cold trap	216.077 kW upper bound	Complete FD packages	Separate from process PG
CIP initial stock	638.611 kW over 2 h	700 kW selected heater	Makeup/startup lower bound below
CIP hot makeup	229.900 kW	Same selected heater	476,563.542 kWh

The exact process PG peak is the overlap of E-501 at 170.630 kW with one J-301 at 317.553 kW. The process chiller excludes upstream feeds already supplied at 4 C, whole-FD packages, CIP and heat leaks. Complete FD procurement already includes refrigeration, cold trap, shelf system, vacuum and controls; these components are counted once. The primary-coldtrap value is a conservative coincidence bound for five active lots, not the whole-FD demand.

9.1 Selected CIP programme and balances

The cleaning design uses four 110 m3/h recirculation circuits and 20 m3 active alkaline stock. A1 wt% NaOH,75 C programme and the 1.5–3m/s pipeline cleaning method are selected using the Tetra Pak handbook as a method analogue. The adopted 75 min group-service allowance remains in the schedule. Cleaning efficacy and detergent endpoints are not inferred from time alone.

There are 13 cleaning services per lot. Selected replacement is 0.5 m3 solution per service:6.5 m3 hot makeup and 29.5 m3 rinse allowance per lot, together 36 m3 hydraulic solution equivalent. At1000 kg/m3 and 1 wt% NaOH the per-lot component input is 35.935 m3 water plus 65 kg NaOH.

Initial/final 20 m3 stock is included in the annual ledger. The annual component input/output is 41165.375 m3 water plus 74625 kg NaOH, with zero calculated CIP mass residual.

The 700 kW heater covers the calculated 638.611 kW initial-stock requirement and 229.900 kW peak makeup duty. Annual 476563.542 kWh thermal is a makeup/startup lower bound; cold metal, return reheating and losses remain to be allocated. It is not added blindly to the old 80 kWh/native-batch generic auxiliary allowance.

9.2 Collection receiver capacities

Collection	Nominal m3	Outlet interface m3/h	Sizing basis
Alkaline and selected CIP	25	40	Cumulative event buffer with 20 percent margin and 80 percent working fill
Acid salt and wash filtrate	35	10	Fast 90 m3/h brine discharge and individual filtrate-slug bound
FD condensate	2	1	1.5 m3 calculated requirement rounded to 2 m3
Wet solid residue	Mass capacity 6t/day	External collection	Maximum rolling 24 h receipt 4.767t; no invented bulk density

L108 brine and L110 wash-filtrate collection terminate at the sameT703 acid/salt receiver. T702 is removed from the new physical-node register. T705 is the separate FD condensate receiver. The 40/10/1 m3/h outlets are receiving-facility interfaces; no final outlet pump is claimed without its receiving head. Collection does not calculate neutralization, pH or treatment performance.

10 Operation control and physical arrangement

Loop or valve	Equipment	Controlled quantity	Setpoint or state	Operating action
TIC-201	T-201	Extraction product temperature	21 C	Modulate 25 C hot-water bypass/flow; count conditioning before 36 hhold
TIC-301	T-301	Salting solution temperature	4 C	Modulate glycol bypass while retaining minimum branch circulation
TIC-501	E-501/T-501	Cold wash-water outlet	4 C	Control glycol valve; divert water to buffer until target reached
FQIC-501	F-301 bank	Individual wash dose	Saved per-round mass/14 modules	After CIP drain warm solution, cold-rinse and refill the header within the group rinse allowance. Start dose totalization only after the 4 C temperature permissive; balance or meter module branches.
PIC-301	F-301	Cake filtration pressure difference	100 kPa	Gas regulator maintains differential with inlet/outlet liquid valves in filtration state

Loop or valve	Equipment	Controlled quantity	Setpoint or state	Operating action
XV-301V	F-301	Vent isolation	Open during fill/unload; closed for filtration	Show an actual valve and destination; verify zero gauge pressure before opening equipment
LAL-501	Cold water/pump sources	Minimum liquid level	1.2m selected minimum surface datum	Stop transfer before dry running; elevated low-point allows batch draining without hiding a heel
FIC-801	CIP	Circulation	110 m3/h per active circuit	Recirculate; branch balancing achieves 1.5m/s in maximum-bore line
TIC-801	CIP	Alkaline supply temperature	75 C project programme	Cleaning timer condition uses achieved return temperature, not heater nameplate alone
CIC-801	CIP	Detergent return concentration	1 wt% NaOH selected	Use calibrated concentration/conductivity before declaring selected programme complete
LSH-701/703/705	Waste receivers	Available volume	Calculated working capacity	Pause discharge when receiving capacity is unavailable; send categories to designated collection

The filter sequence distinguishes vented fill, closed pressure filtration, depressurization, vented discharge and isolated CIP. The next P&ID revision must show a real operable vent valve and destination, gas pressure control, branch-dose measurement, filtrate path, isolation and receiver permissives. Gas-driving pressure is separate from the upstream liquid transfer service.

The 100x72m envelope is a preliminary process allocation used for consistent routed coordinates. Each circular filter-area equivalent is approximately 4.43m diameter;6m allocation cells give 42x12m per 14-module bank and 42x28m for two banks including a4m aisle. This explains why the old 36x20m sketch cannot be treated as the industrial arrangement. Vendor cabinet/filter footprints, tray handling, cold storage and external dispatch allocation remain to be finalized. A specific real site has not been selected.

The site interface requires about 5.57 kt/y wet fish-skin supply, supplied-cold process water or an additional feed-chilling design, industrial wastewater reception, and electrical capacity for process cooling, complete dryers and CIP. These are site requirements rather than claims of existing host-factory spare capacity.

11 Capital reconciliation and economic scenarios

Change from fixed-process V019	Old quantity	New quantity	Purchase-budget change EUR
Salting vessel	4	5	302,518
CIP package	2	4	509,807
Additional 16 m2 complete freeze dryer	0	7	1,750,000

The CIP line is a conservative scenario using one inherited package budget for each independent circuit; this does not assert a supplier skid topology or quotation. The staffing allowance be-

comes 13FTE by the inherited rule. Process utility, pump-bank, building/site and cleaning energy/chemical coverage must be reconciled before calling this a complete final installed budget.

Budget category	Amount
Purchased equipment scenario	28.960 million EUR
Fixed capital with Lang 3.63 once	105.125 million EUR
Initial cash including inherited allowances	119.091 million EUR
NPV-zero powder price within this scenario	142.132EUR/kg powder

Powder price EUR/kg	Annual net profit million EUR	NPV20y10 percent million EUR
50	-7.722	-144.191
100	2.372	-59.702
150	10.876	11.149
200	19.380	82.000
250	27.884	152.852

These results are conditional budget calculations. The inherited model uses 20 years,10 percent discount,20.6 percent tax,80 percent first-year commercial output, working capital and terminal recovery. The 80 percent commercial ramp is separate from the empty-plant physical startup schedule. EUR 250/kg is a scenario per kilogram of powder, not per kilogram of contained collagen or a confirmed annual order. No process-performance savings are credited.

12 Verification and remaining closeout

Model hashes match the frozen originals. The normalized inventory and physical 18x table were compared independently, including 840 scaled mass entries with unchanged streamT/P. The scheduler checks explicit reservations and deadlines. Thermal/hydraulic work includes 60 base checks,23 independent Colebrook comparisons and published-curve candidate screens. Excel provides live capacity and hydraulic/thermal formulas with independent comparisons; these implementation checks do not constitute fish-collagen experiments.

Status	Remaining item	Completion condition
Pending drawing issue	Black-and-white publication figures and revised P&ID/layout	Execute controlled Python scripts in Spyder, visually inspect vector outputs, and review provenance. Drawings are not issued in this revision.
Assigned separately	BFD and PFD	Receive drawings from user assigned team and cross-checktags/streams/boundaries.
Procurement qualification	Minimum flow, materials, seals, offered motors, thin-cake filtration and FD loading	Equivalent-duty supplier data for the issued service requirements.
Engineering reconciliation	CIP heat and chemicals, utility/site cost coverage, and operators	Close resource, energy and cost boundaries without double counting allowances.
Reference comparison	Complete same-output Dialysis architecture/economics	Add discreteFD/referenceequipmentcounts before full-route investment comparison.
Application qualification	Film/coating function and buyer/storage specification	Keep currently unassigned functional data separate from model composition screens.

The current evidence supports a coordinated course-design case and a conditional route recommendation: Washing materially reduces process water and water sent to drying under the adopted endpoints. The next issue should close drawings, procurement and cost interfaces using this controlled basis. The frozen Aspen design does not need additional blocks merely for presentation complexity.

13 References and traceability

[1] Nilsuwan et al.(2025). Properties and Characteristics of Film from Salmon Skin Acid-Soluble Collagen Solution as Influenced by Ultrasonication Process. Foods 14,1088. https://doi.org/10.3390/foods14071088.

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The machine-readable source ledger records 546 hydraulic/thermal parameter rows with sources, locators, classifications, formulas and applicability. Product, capacity, cleaning and collection sources are retained in their corresponding JSONs. Literature values, manufacturer curves, transferred analogues, project choices and calculated results remain distinct.

A Master equipment and service register

Tag or range	Service	Quantity	Capacity per item	Route status
T-101A/B	Alkaline pretreatment and skin rinsing	2	45.000 m3 nominal	Main physical engineering case
T-102	Acetic acid preparation	1	18.000 m3 nominal	Main physical engineering case
T-201A-F	Collagen extraction	6	36.000 m3 nominal	Main physical engineering case
T-301A-E	Salting and cooling	5	36.000 m3 nominal	Main physical engineering case
S-101	Raw skin mechanical preparation	1	4,864.865 kg charge	Main physical engineering case
S-201	Extraction clarification	1	21,462.645 kg batch	Main physical engineering case
S-301	Initial salt separation	1	23,485.578 kg batch	Main physical engineering case
F-301A01-A14 and B01-B14	Eight-cycle cake reslurry washing	28	15.429 m2 active per module	Main physical engineering case
T-501A/B	Metered cold wash-water storage	2	25.000 m3 nominal	Main physical engineering case
E-501	Wash-water cooler	1	24.429 m2 outside tube area	Main physical engineering case
E-401	Redissolved collagen cooler	1	27.483 m2 outside tube area	Reference route thermal case; not installed with Washing
E-402	Dialysis bath make-up cooler	1	146.574 m2 outside tube area	Reference route thermal case; not installed with Washing
FD-G01-A	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G01-B	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G01-C	Freeze drying of one dedicated product lot	1	16.000 m2 active shelves	Main physical engineering case
FD-G02-A	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G02-B	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G02-C	Freeze drying of one dedicated product lot	1	16.000 m2 active shelves	Main physical engineering case
FD-G03-A	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G03-B	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G03-C	Freeze drying of one dedicated product lot	1	16.000 m2 active shelves	Main physical engineering case
FD-G04-A	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G04-B	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case

Tag or range	Service	Quantity	Capacity per item	Route status
FD-G04-C	Freeze drying of one dedicated product lot	1	16.000 m2 active shelves	Main physical engineering case
FD-G05-A	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G05-B	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G05-C	Freeze drying of one dedicated product lot	1	16.000 m2 active shelves	Main physical engineering case
FD-G06-A	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G06-B	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G06-C	Freeze drying of one dedicated product lot	1	16.000 m2 active shelves	Main physical engineering case
FD-G07-A	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G07-B	Freeze drying of one dedicated product lot	1	136.000 m2 active shelves	Main physical engineering case
FD-G07-C	Freeze drying of one dedicated product lot	1	16.000 m2 active shelves	Main physical engineering case
P-701	Alkaline waste	1	90.000 m3/h	Main physical engineering case
P-101	Alkali feed	1	90.000 m3/h	Main physical engineering case
P-501A/B	Cold wash water	2	90.000 m3/h	Main physical engineering case
P-103	Extraction acid	1	90.000 m3/h	Main physical engineering case
P-102	Fresh rinse water	1	90.000 m3/h	Main physical engineering case
P-201A-C	Product slurry transfer	3	90.000 m3/h	Main physical engineering case
P-500	Wash water cooling preparation	1	9.578 m3/h	Main physical engineering case
PG-501	E501 glycol branch circulation	1	78.087 m3/h	Main physical engineering case
PG-301	Salting coil glycol circulation	1	144.000 m3/h	Main physical engineering case
PH-201	Extraction hot-water circulation	1	36.000 m3/h	Main physical engineering case
CH-601	Process-only glycol cooling	1	600.000 kW refrigeration	Main physical engineering case
HW-602	Extraction hot-water service	1	140.000 kW heater	Main physical engineering case
CIP-801	Cleaning service availability	4	110.000 m3/h per circuit	Main physical engineering case
TK-601	Process glycol hydraulic buffer	1	24.000 m3 nominal	Main physical engineering case
EV-601	Glycol expansion vessel	1	3.200 m3 nominal	Main physical engineering case
TK-602	Extraction hot-water buffer	1	4.000 m3 nominal	Main physical engineering case
TK-801	Active CIP alkaline stock	1	25.000 m3 nominal	Main physical engineering case

Tag or range	Service	Quantity	Capacity per item	Route status
T-701	Alkaline and selected CIP collection	1	25.000 m3 nominal	Main physical engineering case
T-703	Combined acid salt and wash-filtrate collection	1	35.000 m3 nominal	Main physical engineering case
T-705	FD condensate collection	1	2.000 m3 nominal	Main physical engineering case
HT-801	CIP initial stock and makeup heating	1	700.000 kW thermal	Main physical engineering case

B Individual equipment data sheets

B.1 T-101A/B Alkaline pretreatment and skin rinsing

Field	Specified requirement or calculation
Type	Agitated vented batch vessel
Quantity	2
Required capacity	45.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	PRETREAT, DRAIN, SKWASH
batch mass kg	29,189.189160
density kg m3	1,000.000000
working liquid m3	29.189189
maximum fill fraction	0.800000
coil displacement m3	0.000000
minimum nominal m3	36.486486
trial cylinder diameter m	3.532
trial cylinder straight height m	4.592
operating pressure bar abs	1.000
cleaning temperature requirement C	75.000
maximum actual reservation h	13.508
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.2 T-102 Acetic acid preparation

Field	Specified requirement or calculation
Type	Agitated vented batch vessel
Quantity	1
Required capacity	18.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	ACIDMIX, ACIDCOOL
batch mass kg	14,308.430184
density kg m3	1,000.000000
working liquid m3	14.308430
maximum fill fraction	0.800000
coil displacement m3	0.000000
minimum nominal m3	17.885538
trial cylinder diameter m	2.603
trial cylinder straight height m	3.383
operating pressure bar abs	1.000
cleaning temperature requirement C	75.000
maximum actual reservation h	3.686
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.3 T-201A-F Collagen extraction

Field	Specified requirement or calculation
Type	Agitated vented batch vessel
Quantity	6
Required capacity	36.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	EXTMIX, EXTRACT
batch mass kg	21,462.645240
density kg m3	1,000.000000
working liquid m3	21.462645
maximum fill fraction	0.800000
coil displacement m3	0.907677
minimum nominal m3	27.962902
trial cylinder diameter m	3.279
trial cylinder straight height m	4.263
operating pressure bar abs	1.000
cleaning temperature requirement C	75.000
maximum actual reservation h	39.940
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.4 T-301A-E Salting and cooling

Field	Specified requirement or calculation
Type	Agitated vented batch vessel
Quantity	5
Required capacity	36.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	SALTMIX, CHILL
batch mass kg	23,485.577580
density kg m3	1,000.000000
working liquid m3	23.485578
maximum fill fraction	0.800000
coil displacement m3	0.993073
minimum nominal m3	30.598313
trial cylinder diameter m	3.279
trial cylinder straight height m	4.263
operating pressure bar abs	1.000
cleaning temperature requirement C	75.000
maximum actual reservation h	28.971
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.5 S-101 Raw skin mechanical preparation

Field	Specified requirement or calculation
Type	Mechanical preparation package
Quantity	1
Required capacity	4,864.865 kg charge
Route and status	Washing; Main physical engineering case
Native/physical mapping	SKIN feed interface
skin kg	4,864.864860
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.6 S-201 Extraction clarification

Field	Specified requirement or calculation
Type	Solid liquid separation package
Quantity	1
Required capacity	21,462.645 kg batch
Route and status	Washing; Main physical engineering case
Native/physical mapping	CLARIFY
inlet mass kg	21,462.645240
residue kg	1,191.808688
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.7 S-301 Initial salt separation

Field	Specified requirement or calculation
Type	Separate initial precipitation separation package
Quantity	1
Required capacity	23,485.578 kg batch
Route and status	Washing; Main physical engineering case
Native/physical mapping	SALTOUT
inlet mass kg	23,485.577580
cake kg	1,210.416534
brine kg	22,275.161100
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.8 F-301A01-A14 and B01-B14 Eight-cycle cake reslurry washing

Field	Specified requirement or calculation
Type	Two independently scheduled banks of 14 pressure-filter modules
Quantity	28
Required capacity	15.429 m2 active per module
Route and status	Washing; Main physical engineering case
Native/physical mapping	CAKEWASH
banks	2.000000
modules per bank	14.000000
active area per bank m2	216.000000
nominal area upper bound m2	16.000000
first reslurry m3	3.631250
first reslurry per module m3	0.259375
wash water m3 per lot	19.155806
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.9 T-501A/B Metered cold wash-water storage

Field	Specified requirement or calculation
Type	Insulated local buffer
Quantity	2
Required capacity	25.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	PWATER/PWAT4
wash water m3	19.155806
maximum fill fraction	0.800000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.10 E-501 Wash-water cooler

Field	Specified requirement or calculation
Type	Countercurrent assembled parallel annular bank
Quantity	1
Required capacity	24.429 m2 outside tube area
Route and status	Washing; Main physical engineering case
Native/physical mapping	WATCOOL
physical batch scale	18.000000
batch mass kg	19,155.806100
batch energy kWh	341.260780
duty kW	170.630390
conditioning h	2.000000
process flow m3 h	9.577903
glycol flow m3 h	78.086904
U W m2 K	808.289626
LMTD K	11.628170
process dp kPa	23.042206
glycol dp kPa	210.913974
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.11 E-401 Redissolved collagen cooler

Field	Specified requirement or calculation
Type	Countercurrent assembled parallel annular bank
Quantity	1
Required capacity	27.483 m2 outside tube area
Route and status	Dialysis; Reference route thermal case; not installed with Washing
Native/physical mapping	REDCHILL
physical batch scale	18.000000
batch mass kg	19,038.363480
batch energy kWh	312.445438
duty kW	156.222719
conditioning h	2.000000
process flow m3 h	9.519182
glycol flow m3 h	71.493410
U W m2 K	629.977445
LMTD K	11.329819
process dp kPa	27.982269
glycol dp kPa	201.765696
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.12 E-402 Dialysis bath make-up cooler

Field	Specified requirement or calculation
Type	Countercurrent assembled parallel annular bank
Quantity	1
Required capacity	146.574 m2 outside tube area
Route and status	Dialysis; Reference route thermal case; not installed with Washing
Native/physical mapping	BATHCOOL
physical batch scale	18.000000
batch mass kg	371,844.882000
batch energy kWh	6,613.545150
duty kW	1,102.257525
conditioning h	6.000000
process flow m3 h	61.974147
glycol flow m3 h	504.434629
U W m2 K	852.065044
LMTD K	11.628170
process dp kPa	26.364673
glycol dp kPa	239.839601
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.13 P-701 Alkaline waste

Field	Specified requirement or calculation
Type	Compatible liquid transfer channel
Quantity	1
Required capacity	90.000 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical transfer; native block inventories unchanged
selected channels	1.000000
fixed earliest cleaning minimum channels	1.000000
min gap h	0.520270
flow each m3 h	90.000000
post transfer CIP h	0.000000
sharing basis	Fixed compatibility group and no queuing; curves/motor provided in hydraulic register
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.14 P-101 Alkali feed

Field	Specified requirement or calculation
Type	Compatible liquid transfer channel
Quantity	1
Required capacity	90.000 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical transfer; native block inventories unchanged
selected channels	1.000000
fixed earliest cleaning minimum channels	1.000000
min gap h	6.534996
flow each m3 h	90.000000
post transfer CIP h	0.000000
sharing basis	Fixed compatibility group and no queuing; curves/motor provided in hydraulic register
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.15 P-501A/B Cold wash water

Field	Specified requirement or calculation
Type	Compatible liquid transfer channel
Quantity	2
Required capacity	90.000 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical transfer; native block inventories unchanged
selected channels	2.000000
fixed earliest cleaning minimum channels	2.000000
min gap h	0.031841
flow each m3 h	90.000000
post transfer CIP h	0.000000
sharing basis	Fixed compatibility group and no queuing; curves/motor provided in hydraulic register
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.16 P-103 Extraction acid

Field	Specified requirement or calculation
Type	Compatible liquid transfer channel
Quantity	1
Required capacity	90.000 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical transfer; native block inventories unchanged
selected channels	1.000000
fixed earliest cleaning minimum channels	1.000000
min gap h	6.646284
flow each m3 h	90.000000
post transfer CIP h	0.000000
sharing basis	Fixed compatibility group and no queuing; curves/motor provided in hydraulic register
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.17 P-102 Fresh rinse water

Field	Specified requirement or calculation
Type	Compatible liquid transfer channel
Quantity	1
Required capacity	90.000 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical transfer; native block inventories unchanged
selected channels	1.000000
fixed earliest cleaning minimum channels	1.000000
min gap h	0.520271
flow each m3 h	90.000000
post transfer CIP h	0.000000
sharing basis	Fixed compatibility group and no queuing; curves/motor provided in hydraulic register
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.18 P-201A-C Product slurry transfer

Field	Specified requirement or calculation
Type	Compatible liquid transfer channel
Quantity	3
Required capacity	90.000 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical transfer; native block inventories unchanged
selected channels	3.000000
fixed earliest cleaning minimum channels	3.000000
min gap h	4.801306
flow each m3 h	90.000000
post transfer CIP h	1.250000
sharing basis	Fixed compatibility group and no queuing; curves/motor provided in hydraulic register
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.19 P-500 Wash water cooling preparation

Field	Specified requirement or calculation
Type	Utility circulation pump
Quantity	1
Required capacity	9.578 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External mechanical service
line	L500
head m	7.500000
NPSHa m	10.304570
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.20 PG-501 E501 glycol branch circulation

Field	Specified requirement or calculation
Type	Utility circulation pump
Quantity	1
Required capacity	78.087 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External mechanical service
line	G501
head m	36.000000
NPSHa m	10.191613
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.21 PG-301 Salting coil glycol circulation

Field	Specified requirement or calculation
Type	Utility circulation pump
Quantity	1
Required capacity	144.000 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External mechanical service
line	G301
head m	9.500000
NPSHa m	10.179403
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.22 PH-201 Extraction hot-water circulation

Field	Specified requirement or calculation
Type	Utility circulation pump
Quantity	1
Required capacity	36.000 m3/h
Route and status	Washing; Main physical engineering case
Native/physical mapping	External mechanical service
line	H201
head m	9.500000
NPSHa m	10.309752
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.23 CH-601 Process-only glycol cooling

Field	Specified requirement or calculation
Type	Process chiller package
Quantity	1
Required capacity	600.000 kW refrigeration
Route and status	Washing; Main physical engineering case
Native/physical mapping	External utility
glycol concentration volpct	30.000000
supply C	-2.000000
return C	0.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.24 HW-602 Extraction hot-water service

Field	Specified requirement or calculation
Type	Closed hot-water conditioning loop
Quantity	1
Required capacity	140.000 kW heater
Route and status	Washing; Main physical engineering case
Native/physical mapping	External utility
supply C	25.000000
flow m3 h	36.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.25 CIP-801 Cleaning service availability

Field	Specified requirement or calculation
Type	Independently dispatched recirculation circuits
Quantity	4
Required capacity	110.000 m3/h per circuit
Route and status	Washing; Main physical engineering case
Native/physical mapping	External cleaning system
selected group service circuits	4.000000
immediate cleaning fixed clock circuits	7.000000
dispatch method	One dedicated FD group service circuit; general CIP uses earliest-deadline dispatch before equipment reuse. Material operations are unchanged.
maximum simultaneous nozzle branches	16.000000
branch peak at h	91.529487
group service CIP h per lot	16.250000
whole plant fresh CIP water allowance m3 per lot	36.000000
water boundary	36 m3/lot is the inherited whole-plant allowance; it is not 36 m3 for every equipment CIP event.
topology	A filter-bank service includes 14 parallel module branches; an FD-group service includes 3 cabinet branches. Circuit flow/return capacity must support these groups.
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.26 TK-601 Process glycol hydraulic buffer

Field	Specified requirement or calculation
Type	Utility or collection receiver
Quantity	1
Required capacity	24.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical interface
selected nominal m3	24.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.27 EV-601 Glycol expansion vessel

Field	Specified requirement or calculation
Type	Utility or collection receiver
Quantity	1
Required capacity	3.200 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical interface
selected nominal m3	3.200000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.28 TK-602 Extraction hot-water buffer

Field	Specified requirement or calculation
Type	Utility or collection receiver
Quantity	1
Required capacity	4.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical interface
selected nominal m3	4.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.29 TK-801 Active CIP alkaline stock

Field	Specified requirement or calculation
Type	Utility or collection receiver
Quantity	1
Required capacity	25.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical interface
selected nominal m3	25.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.30 T-701 Alkaline and selected CIP collection

Field	Specified requirement or calculation
Type	Utility or collection receiver
Quantity	1
Required capacity	25.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical interface
selected nominal m3	25.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.31 T-703 Combined acid salt and wash-filtrate collection

Field	Specified requirement or calculation
Type	Utility or collection receiver
Quantity	1
Required capacity	35.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical interface
selected nominal m3	35.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.32 T-705 FD condensate collection

Field	Specified requirement or calculation
Type	Utility or collection receiver
Quantity	1
Required capacity	2.000 m3 nominal
Route and status	Washing; Main physical engineering case
Native/physical mapping	External physical interface
selected nominal m3	2.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.33 HT-801 CIP initial stock and makeup heating

Field	Specified requirement or calculation
Type	CIP heater requirement
Quantity	1
Required capacity	700.000 kW thermal
Route and status	Washing; Main physical engineering case
Native/physical mapping	External utility
total active solution m3	20.000000
parallel loop allowance	4.000000
solution m3 per loop	5.000000
NaOH mass fraction	0.010000
initial NaOH kg	200.000000
selected temperature C	75.000000
cp kJ kg K	4.180000
density kg m3	1,000.000000
Reference	V009 fixed inventory; V021 dispatch; V021 hydraulic/thermal calculation; D21 physical equipment choice
Wetted material basis	316L product contact is the trial selection; confirm chloride/acid and cleaning compatibility for procurement.

B.34 Freeze dryer individual cabinet loading

Cabinet	Active area m2	Wet feed kg	Water kg	Primary capacity kg/h
FD-G01-A	136	557.599	473.600	25.500
FD-G01-B	136	557.599	473.600	25.500
FD-G01-C	16	65.600	55.718	3.000
FD-G02-A	136	557.599	473.600	25.500
FD-G02-B	136	557.599	473.600	25.500
FD-G02-C	16	65.600	55.718	3.000
FD-G03-A	136	557.599	473.600	25.500
FD-G03-B	136	557.599	473.600	25.500
FD-G03-C	16	65.600	55.718	3.000
FD-G04-A	136	557.599	473.600	25.500
FD-G04-B	136	557.599	473.600	25.500
FD-G04-C	16	65.600	55.718	3.000
FD-G05-A	136	557.599	473.600	25.500
FD-G05-B	136	557.599	473.600	25.500
FD-G05-C	16	65.600	55.718	3.000
FD-G06-A	136	557.599	473.600	25.500
FD-G06-B	136	557.599	473.600	25.500
FD-G06-C	16	65.600	55.718	3.000
FD-G07-A	136	557.599	473.600	25.500
FD-G07-B	136	557.599	473.600	25.500
FD-G07-C	16	65.600	55.718	3.000

Every listed cabinet has the same 43.917873 h cycle requirement and 5 Pa chamber pressure basis. The complete package includes its cold trap, refrigeration, shelf system, vacuum and controls. The large/small loadings preserve the accepted per-area model assumptions.